

CHEM 101 Topic 4: Atomic Structure — Problem Set

Instructions: Complete the following problems using the analysis method practiced in your guided assignment. Show all your work, including your reasoning for the atomic models and your step-by-step filling of electron configurations.

Section 1: History of Atomic Models

Problem 1: Contrast the "Plum Pudding" model of J.J. Thomson with the model proposed by Ernest Rutherford. What was the key discovery that led to the change?

Problem 2: Why is the Quantum Model more accurate than the Bohr model when describing the location of an electron?

Problem 3: Which early scientist proposed that the atom was a solid, indivisible sphere, and on what basis did he create this theory?

Section 2: Atomic Notation and Mass

Problem 4: An atom has an atomic number of 17 and a mass number of 35. Identify the element and determine the number of protons, neutrons, and electrons in a neutral atom of this element.

Problem 5: Chlorine has two common isotopes: Chlorine-35 and Chlorine-37. How do these two isotopes differ in terms of their subatomic particles?

Problem 6: Calculate the mass number of an isotope that contains 26 protons and 30 neutrons.

Section 3: Energy Levels & Orbitals

Problem 7: What is the maximum number of electrons that can occupy the second energy level ($n=2$) in total? Break this down by sublevel (s and p).

Problem 8: If an electron is in a "3d" orbital, what is its energy level and what is its sublevel?

Problem 9: Explain why an electron cannot enter the 2p sublevel until the 2s sublevel is completely filled. Which principle explains this?

Section 4: Electron Configurations

Problem 10: Write the full electron configuration for Magnesium (Mg), which has an atomic number of 12.

Problem 11: Write the full electron configuration for Sulfur (S), which has an atomic number of 16.

Problem 12: Write the electron configuration for a Calcium ion (Ca^{2+}), given that neutral Calcium has an atomic number of 20.